CSC101 Assignment

Data Exploration & Collection

Overview

Data is everywhere! Computers are used in an iterative and interactive way to process information and help us gain insight and knowledge. Patterns can emerge when data is collected and transformed using computational tools. In this assignment, you will use the data science process to communicate your own data story!

Assignment Preparation

Make a copy of this document to record your answers. Rename your copy of the document to **lastname_data**, where **lastname** is your last name. You will **place all answers and screenshots** in this Google Document. The steps below will tell you what to put in your answer document - use the spaces provided to insert your answers accordingly.

Part 1: Data Exploration

We collect data to learn about the world and test our hypotheses about the patterns and relationships that might otherwise be invisible. Computational tools now make it possible to quickly collect and analyze vast amounts of data, but those processes still depend on knowledgeable and thoughtful people who understand the data being used and can clearly present the conclusions drawn from it.

Data storytelling is an effective way to present data that can lead to actionable insight and improved decision making. A *data story* communicates data analysis and insights using data, visuals, and narrative that allows the audience to understand its meaning and draw conclusions from it.

DO THE FOLLOWING:

- ➔ Watch: Dear Data [3:18]
- ➔ Select one of the Dear Data postcard examples to explore in depth:

Complaints	Laughter	Phone Addiction	Beauty

- ➔ Respond to the following question prompts:

Which example did you choose to explore?	

What does the visualization communicate?	
What is a specific pattern that you noticed in the visual?	
How did you know how to read the visual?	
What conclusions can you make about the authors' experience?	
What sources of variability are represented in each visual?	

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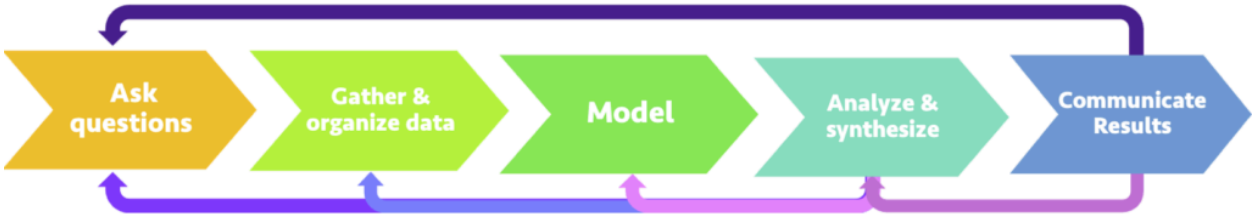
Part 2: Dear Data Postcard

In this part of the assignment you will use the data science process to create your own Dear Data postcard - see student examples. The data science process is an iterative process that includes feedback, revision, and refinement at every stage.



➢ Ask Questions ➢

In this stage, you will form a question for your own Dear Data postcard.

🔍 Review the student examples and consider other topic ideas to help you decide on a question or topic for your postcard (you may form your own question or use one from the examples).

➔ Answer the question prompt below:

What question or topic will you explore?	

➢ Gather & Organize Data ➢

In this stage, you will form a data gathering plan and collect the data necessary to create your Dear Data postcard.

🔍 Review the [examples]() to help you form a data gathering plan.

➔ Answer the question prompts to explain how will you gather your data:

Data Gathering Plan	
How will you define what data you are gathering?	
How much data will you need to gather?	
In order to gather enough data, how long will you need to gather data: all day, a few hours? *Limit your data collection period to one day or less so you have plenty of time to work through the iteration process.*	
How will you record your data?	
What variability might you see in your data?	

➔ Once you have a plan in place, you can begin to gather data

➔ Record the data that will support your topic using the method and time period determined in your plan above

➔ After the data collection period has ended, consider what worked well and what didn't with your data gathering

➔ Answer the question prompts to determine how to revise your plan:

Plan Revision	
How will you change your data gathering plan based on your experience?	
Will you change your question? What will your new question be?	
How will you change how you define what you are gathering?	
Do you need to change the amount of time you are gathering data over? How?	

➔ Revise your plan, as necessary. Depending on how your data gathering has gone, these revisions may be very minimal or more major. It may involve gathering some additional data or it may entail revising your question significantly.

➢ Model ➢

In this stage, you will create a model of your data, in this case a visual representation, that allows us to explore patterns in the data.

🔍 Review the student examples for visual creation ideas - this is an opportunity to be creative in your design!

➔ Answer the question prompts to help guide the direction of your visual:

Data Visual Planning	
What did you find interesting about your data?	
What are the factors you will need to show in your data?	
What sources of variability are in your data? How might you represent that variability in your visual?	

What might your key look like? Are there specific types of visuals you are drawn to?	

- Create a draft of your postcard visual. Consider the following:
  - You may wish to start drawing and then create a key or create a key first and then begin your drawing - either approach is ok.
  - You may wish to draw on paper or create an electronic drawing using a tablet or computer.
  - You may create multiple drafts before you are satisfied with the result - this is normal and part of the process.
- Share your draft with someone (classmate, friend, family, teacher, etc.) to get feedback.
  - Ask for feedback, suggestions, or questions they have that might need clarification.
- Create a final draft of your data visual based on feedback.
  - Your visual must have a key that clearly communicates how to read and translate the visual
- Insert an image of your final Dear Data visual.

[Insert image here]

Analyze and Synthesize

In this stage, you will analyze your data visual and reflect on the data science process.

- In 3-5 sentences each, answer the following question prompts to help analyze your data.

Summative Analysis	
Provide a description of your visual.	
What did you learn from your visual about your data?	

- Answer the following question prompts to reflect on the data science process:

Reflection	
What did you learn from the process of collecting data?	
What worked well? What didn't work?	
What would you do differently next time?	
What did you learn while creating the visual?	
What did you learn about yourself from the project?	

➢ Communicate Results ➢

In this stage, you will create slides to communicate what you have learned and created throughout the data science process.

- ➔ Create a new Google Slide presentation to communicate your results
- ➔ Add the following to your slides:
  - A slide containing your question and your plan for collecting data
  - A picture of your collected data (doesn't need to be cleaned up)
  - Your final draft of your data visual with a key that clearly communicates how to read and translate the visual
  - Your summative analysis of your data visual
  - Your reflection on the data science process
- ➔ Provide the URL of your Google Slides presentation. Be sure that the permission is set so that anyone with the link can view it.

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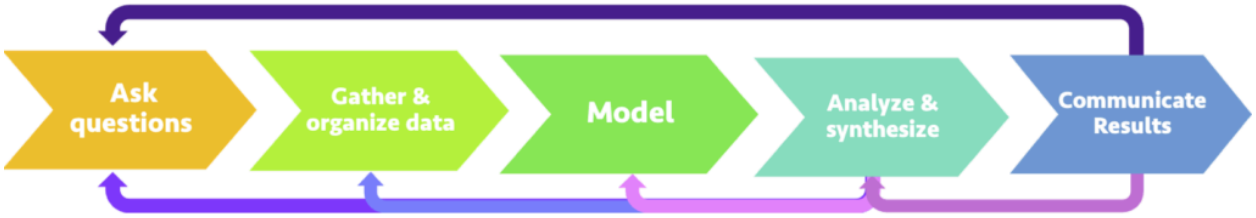
Part 3: Computing Innovations Data Collection

In this part of the assignment you will use the data science process to begin collecting data based on your computing innovations project topic.



➢ Ask Questions ➢

For the purpose of this assignment, you will start with a broad question:

- ➔ "What impact does [TOPIC] have on society?" - you will refine your question in a later assignment using the data science process.

➢ Gather & Organize Data ➢

In this stage, you will use Google Forms to collect original data about your final project topic. You will create a survey about the impact of your topic on society and gather at least 20 responses in a spreadsheet for future analysis.

- 🔍 Review the example based on the *Impact of Email* to help design your survey.
- ➔ Follow this tutorial on how to Create a Google form.
- ➔ Give the Form a Title (for example, *The Impact of Email*).
- ➔ Give the Form a brief Description so that people taking your survey know why you are asking them to take it.
- ➔ Create questions following the guidelines below, where [TOPIC] is your project topic:

  Create the question type *exactly* as specified. Note that how to answer some of these questions is left intentionally vague so that we can use the data collected as a candidate for filtering and cleaning in a later lesson.

Question	Question Type
How old are you?	Short answer

	Short answer
What is the gender you most closely identify?	Multiple choice (you provide the choices) Multiple choice
Before taking this survey, have you heard of [TOPIC]?	Multiple choice (you provide the choices) Multiple choice
On a scale of 1 (low impact) to 5 (high impact) how much impact do you think [TOPIC] has had on society in general?	Linear scale (1-5) Linear scale
In your opinion, has the Impact of Email on society been: • Mostly Positive • Mostly Negative • Neutral • I'm not sure	Multiple choice (include the options provided) Multiple choice
On a scale of 1 (low impact) to 5 (high impact) how much impact do you think [TOPIC] has had on your life?	Linear scale (1-5) Linear scale
In ten words or less, how has [TOPIC] impacted your life?	Short answer Short answer

You must ask the above questions. You may also add other questions if you wish, but not too many as you don't want your survey to be burdensome.

➔ Check the Form's Response setting to be sure it's not restricted to your school's domain (see example)

Restrict to users in University of Rhode Island and its trusted organizations

- Send the form to at least 20 people (more is better). Be sure to collect at least 20 responses (you will analyze the results in the next assignment).
    - You may wish to text/email your survey to friends and family, post on social media, post on a course discussion board/forum, etc. in an effort to collect enough responses.
    - Be sure to get responses from at least 2 different communities of people (i.e. peers and non-peers, different age groups, different geographic locations, etc.) so that your results include more diverse perspectives.
    - If you are having trouble collecting enough responses after trying the methods above, contact your instructor for further guidance.

Place in your assignment document:

- The URL of your survey.

- In a few sentences, explain how you could further expand the scope of your survey to include responses from different communities and cultures (provide at least one example).